

MuST: Multi-granularity Sub-Text Queries for Few-Shot Action Recognition

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Abstract. Few-shot action recognition with vision–language models typically represents an action class with a single text prompt, which captures class-level semantics but provides limited information about temporal progression. Recent work decomposes a class into atomic phase descriptions, but when these phases are treated as an independent set, the representation remains largely insensitive to their temporal order. We present **MuST**, a **multi-granularity sub-text pyramid** for few-shot action recognition. A language model decomposes each class name offline into N consecutive atomic phases, and MuST constructs all contiguous phase spans, yielding $M = N(N + 1)/2$ prompts that describe the action from individual phases to longer temporal transitions. The span structure further induces a coarse temporal geometry over the clip, which we use to build a **span-aligned attention prior** that softly biases each prompt’s cross-attention towards the frames it is likely to describe. On HMDB51 and Kinetics-100 under the 5-way protocol, MuST improves over the atomic sub-text baseline by 3.8 and 1.9 points at 1-shot, respectively. A control that permutes span centres while keeping span widths fixed supports that the gain comes from temporal alignment rather than generic attention regularisation.

Keywords: Few-shot action recognition · Vision–language models · Temporal modelling.

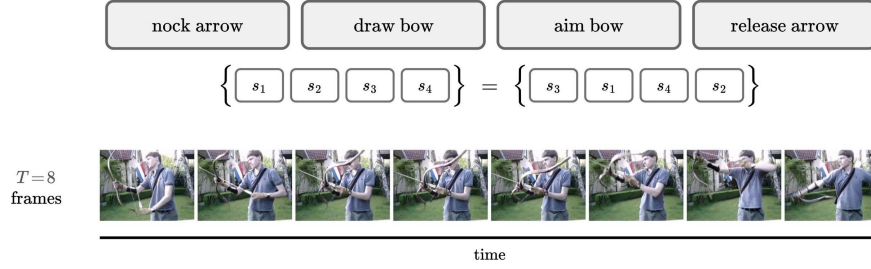
1 Introduction

Few-shot action recognition (FSAR) aims to recognise novel action classes from one or a few labelled videos, commonly through metric-based episodic learning [19,16]. Unlike image recognition, FSAR has an inherent temporal difficulty: two clips of the same action may start at different moments, proceed at different speeds, or allocate different durations to the same sub-action. Most prior methods address this issue on the visual side, for example by temporal alignment or temporal relation modelling [1,10].

Vision–language pre-training offers another source of supervision. CLIP [13] provides a text encoder that can inject class-level semantic knowledge beyond the few available support videos [22]. However, representing a class with a single prompt such as “*a video of a person performing archery*” mainly describes the action category, not

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(a) Atomic sub-texts



(b) Sub-text pyramid

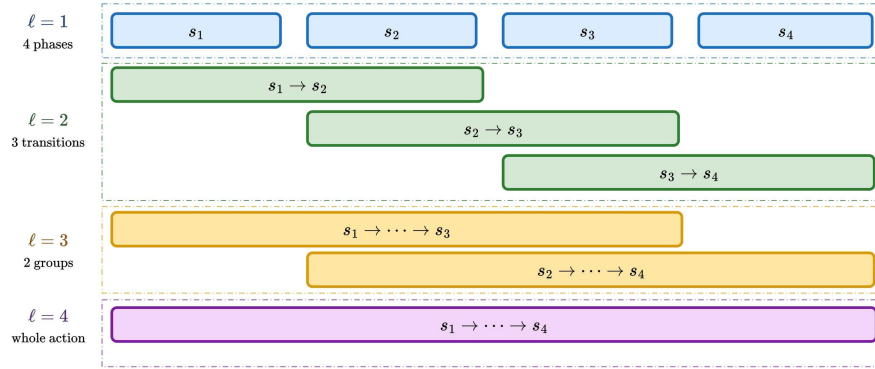


Fig. 1. Construction of the text queries for a class. Given N ordered atomic phases, we concatenate every contiguous span into one sentence, giving $M = N(N + 1)/2$ prompts at multiple temporal granularities. The span indices also provide the centre and width used by the span-aligned attention prior.

its temporal progression. Recent work therefore uses a language model [9] to decompose each class name into N atomic phase descriptions [7,12]. For *Archery*, these may be *nock arrow*, *draw bow*, *aim bow*, and *release arrow*.

Using the atomic phases as an independent set discards part of this information. For example, exchanging *draw bow* and *release arrow* leaves the set of phrases unchanged, although the two orders describe different temporal progressions. The generated list already contains an ordering signal, since the phases are requested in sequence. The key question is how to preserve this order in the class representation and how to use it when matching text queries to video frames.

We address this with **MuST**, a multi-granularity sub-text representation for few-shot action recognition. First, instead of using only the N atomic phrases, we concatenate every contiguous span of them into a single sentence. This yields $M = N(N + 1)/2$ prompts, including single phases, short transitions, longer phase groups, and the full action sequence. Since the text encoder reads an ordered token sequence, prompts such as “*draw bow. Then release arrow.*” and “*release arrow. Then draw bow.*” produce different

embeddings. The resulting pyramid therefore lets the class representation encode phase order at several temporal scales without adding a temporal text module.

Second, the same span structure gives each prompt a coarse temporal extent. Phases i through j are associated with the interval $[(i-1)/N, j/N]$ on the normalised timeline, which defines a centre and a width for the corresponding prompt. We use these values to build an additive bias on the cross-attention logits. Frames inside the expected span are left unchanged, while frames outside it are down-weighted by a soft quadratic penalty. This prior is only possible after forming the pyramid: a global prompt has no span, and atomic prompts provide only fixed-width intervals.

Our contributions are:

- A multi-granularity sub-text pyramid that converts N ordered atomic phases into $M = N(N+1)/2$ contiguous-span prompts, allowing the class representation to preserve phase order at multiple temporal scales.
- A span-aligned attention prior that uses the centre and width of each span to bias cross-attention toward its expected temporal region, while allowing the model to relax the prior during training.

2 Related Work

Few-shot action recognition. Few-shot action recognition commonly extends Matching Networks and Prototypical Networks [19,16] to the video setting. A central difficulty is temporal misalignment: support and query videos of the same class may contain the same action phases at different speeds and positions. Prior methods mainly address this issue from visual features, for example through differentiable temporal alignment [1], cross-transformers over frame tuples [10], cross-video relation modelling [24,17], motion cues [23], or class-agnostic pattern tokens [6]. These approaches differ in how they match videos, but their temporal structure is derived primarily from the visual branch. MuST is complementary: the visual branch includes a fixed motion module that is ablated separately, while the proposed temporal structure comes from class-specific text queries.

Vision–language models for action recognition. Vision–language pre-training has been adapted to video by tuning CLIP-style models with video data or video-specific objectives [21,8,15]. In few-shot action recognition, the text encoder provides class-level information that may be missing from the limited support videos. CLIP-FSAR [22] uses the class prompt to modulate visual prototypes, which corresponds to a single text query per class in our notation. More recent language-guided methods decompose a class name into LLM-generated atomic descriptions [7,12]. When these descriptions are used as an unordered set, they provide fine-grained semantics but do not explicitly preserve phase order. MuST reuses the same offline generation setting, but builds on the ordered list: it forms all contiguous spans as text queries and uses their span indices to define a temporal attention prior.

Multi-scale temporal modelling. Multi-scale temporal representations have been widely used in action recognition, including segment-based sampling [20], temporal relations over frame tuples [27], two-pathway networks operating at different frame rates [3], and temporal feature pyramids [25]. These methods build multi-scale structure on the visual sequence. The first component of MuST instead builds a pyramid on the text side: contiguous spans of ordered phase descriptions are concatenated and encoded by a frozen text encoder. This produces multiple text queries for the same class, each corresponding to a different temporal granularity.

Attention biases and prompting. Additive biases on attention logits are widely used to encode positional structure, from absolute position [18] to relative or recency-based biases [14,11]. Such biases are usually functions of token positions. In MuST, the bias is tied to the span of each text query, giving different queries in the same class different centres and widths. The resulting prior is query-specific rather than only position-specific. Unlike prompt ensembling [13], MuST does not average paraphrases into one class embedding. It matches discrete nested temporal spans to video features individually.

3 Methodology

MuST represents each action class using a collection of text queries constructed from its atomic sub-texts and guides each query towards its expected temporal region in the video. Figure 2 presents an overview of the framework. We follow the standard episodic protocol with disjoint training and test classes. Each episode is formulated as a W -way K -shot task, in which query videos are classified into one of W classes using K labelled support videos per class. Throughout this section, D denotes the embedding dimension of the vision-language model, T the number of sampled frames per video, N the number of atomic sub-texts per class, and M the resulting number of text queries. We use x^q to denote a query video and $x_{c,k}^s$ to denote the k -th support video of class c .

3.1 Multi-granularity Sub-text Pyramid

Sub-text generation. For each class c , we construct an ordered list $S_c = [s_{c,1}, \dots, s_{c,N}]$ of short phrases describing N consecutive action phases. The list is generated once per dataset by prompting a language model [7] with the class name alone, without access to any video. The phrases are retained in temporal order because the span-aligned prior introduced below uses their relative positions within the action sequence.

Pyramid construction. Treating the N phrases as an unordered set would discard their temporal order. We instead retain every contiguous span,

$$\Omega = \{\omega = (i, j) : 1 \leq i \leq j \leq N\}, \quad M = |\Omega| = N(N+1)/2, \quad (1)$$

and render each span as one sentence $\text{Join}(s_{c,i}, \dots, s_{c,j})$ by joining its constituent phrases with the connective *Then*. For $N = 4$, the construction produces $M = 10$ prompts across four levels of increasing span length: four single-phase queries, three

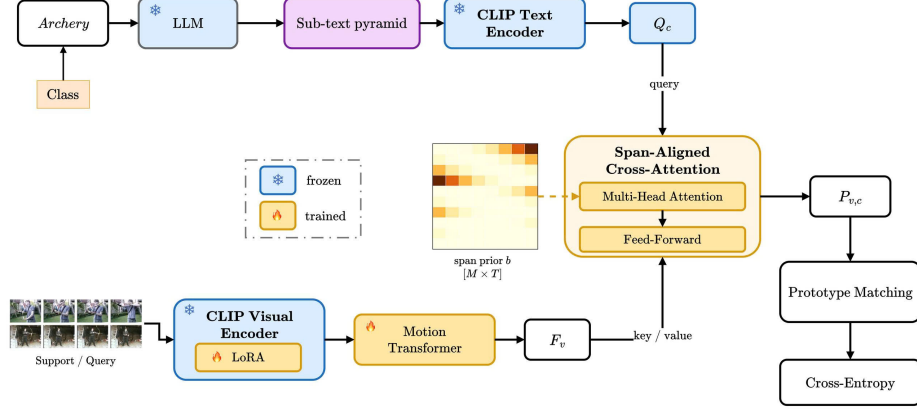


Fig. 2. Overview of MuST. The text branch is encoded once and stored in the class-specific buffer Q_c . The visual branch processes T sampled frames using the visual encoder and motion Transformer. A cross-attention block then combines the text queries with the frame features, with the span-aligned prior added to the attention logits.

two-phase transitions, two three-phase groups, and one whole-action query. Writing ϕ for the text encoder, the queries of class c are

$$Q_c = [q_{c,\omega}]_{\omega \in \Omega} \in \mathbb{R}^{M \times D}, \quad q_{c,\omega} \propto \phi(\text{Join}(s_{c,i}, \dots, s_{c,j})), \quad \|q_{c,\omega}\|_2 = 1. \quad (2)$$

Because the text encoder is frozen, Q_c is computed once during initialisation and stored as a class-specific buffer.

3.2 Visual Branch

CLIP encodes individual frames without explicitly modelling their temporal interactions. We therefore adapt the visual tower using a rank-8 LoRA branch [4] on the query and value projections of every Transformer block, together with a motion Transformer that processes the frame embeddings and their first-order differences. Both adaptation branches are controlled by scalar gates initialised to zero, so the initial representation remains identical to that of the pretrained CLIP visual encoder. The output is ℓ_2 -normalised to obtain $F_v = [f_1, \dots, f_T] \in \mathbb{R}^{T \times D}$. The same LoRA configuration is used throughout the experiments, while the temporal-context module is varied only in its corresponding ablation.

3.3 Span-aligned Cross-attention

Span geometry. A single multi-head cross-attention block allows the text queries to read the frame features F_v . The text representations act as attention queries, while the frame features, augmented with a fixed sinusoidal temporal encoding PE, serve as keys and values.

Standard cross-attention does not explicitly associate a text query with a particular temporal region, and episodic class labels provide no frame-level supervision for learning this association. Because the generated phases are ordered, we assign sub-text k to the k -th portion of the normalised action timeline. Consequently, a span $\omega = (i, j)$ is associated with the interval $[(i-1)/N, j/N]$. Writing μ_ω for its centre, u_ω for its half-width, and p_t for the position of frame t , we define

$$\mu_\omega = \frac{i-1+j}{2N}, \quad u_\omega = \frac{j-i+1}{2N}, \quad p_t = \frac{t-0.5}{T}. \quad (3)$$

This mapping requires neither learned alignment parameters nor additional temporal annotations. A global prompt corresponds only to the full video and therefore yields no local temporal guidance. Ordered atomic queries provide distinct centres but share the same span width, whereas the pyramid contains queries with both different centres and different temporal extents.

Span prior. We introduce a soft penalty on the attention logits for frames outside the interval assigned to each query. The penalty is zero inside the interval and increases quadratically with the distance from its boundary,

$$b[\omega, t] = -g_{|\omega|} \left(\frac{\text{ReLU}(|p_t - \mu_\omega| - u_\omega)}{\delta} \right)^2, \quad (4)$$

where $|\omega| = j - i + 1$ denotes the span length, $g \in \mathbb{R}^N$ contains one gate for each span length, and δ is a shared slope parameter. Both are parameterised through a softplus function to remain positive. The cross-attention block is written as

$$H_{v,c} = Q_c + \text{MHA}(\text{LN}(Q_c), \text{LN}(F_v + \text{PE}), b), \quad P_{v,c} = H_{v,c} + \text{FFN}(\text{LN}(H_{v,c})), \quad (5)$$

where $\text{MHA}(Q, KV, b)$ denotes multi-head attention with Q as the query, KV as the shared key-value sequence, and b added to the attention logits. The output $P_{v,c} \in \mathbb{R}^{M \times D}$ represents video v under the sub-text queries of candidate class c . The resulting bias has size $M \times T$ and is broadcast across the batch and attention heads; it does not require a separate alignment estimate for each video.

For the whole-action span, $|\omega| = N$ gives $\mu_\omega = u_\omega = \frac{1}{2}$, and its row in b is therefore zero. As the corresponding gate approaches zero, the prior for that span vanishes and Eq. (5) reduces to ordinary cross-attention. The prior acts as a soft quadratic penalty rather than a hard mask, so frames outside the expected interval remain accessible to the attention block.

3.4 Prototype Matching

For each class, we average the patterns obtained from its K support videos to form a class prototype. A query video is then read once for each candidate class and scored using the mean cosine similarity across the M granularities,

$$R_c = \frac{1}{K} \sum_{k=1}^K P_{x_{c,k}^s, c}, \quad S(x^q, c) = \tau \cdot \frac{1}{M} \sum_{\omega \in \Omega} \cos(P_{x^q, c}[\omega], R_c[\omega]), \quad (6)$$

where τ is a logit scale. Computing the similarity separately for each span preserves the multi-granularity structure during matching, while averaging the resulting scores aggregates evidence across temporal scales. Training minimises the episodic cross-entropy $\mathcal{L} = \text{CE}(S(x^q, \cdot), y)$, where y denotes the label of the query video. MuST requires no support-set adaptation, auxiliary training objective, or separate global classification branch.

4 Experiments

4.1 Experimental Setup

Datasets and protocol. We evaluate MuST on HMDB51 [5], using the 31/10/10 class split introduced by ARN [26], and on Kinetics-100 [2], using the 64/12/24 split of CMN [28]. We follow the standard 5-way 1-shot and 5-shot protocols. Our results are averaged over 10,000 test episodes with one query per class. The ablation tables additionally report 95% confidence intervals.

Implementation details. We use CLIP ViT-B/16 with an embedding dimension of $D = 512$. The pretrained parameters of both CLIP towers are frozen. Trainable rank-8 LoRA adapters are inserted into the query and value projections of all twelve visual Transformer blocks. Each video is represented by $T = 8$ uniformly sampled frames, with one frame drawn from each temporal segment. We train for 10,000 episodes using AdamW and mixed-precision computation on one RTX 5090. For each dataset, GPT-4 [9] generates one sub-text catalogue from the class names, with $N = 4$ ordered phases per class. All 5-way 1-shot ablations use the same data splits, episode sampler, seed, training schedule, and catalogue.

4.2 Comparison with Existing Methods

As shown in Table 1, MuST achieves the highest accuracy among the compared methods on both datasets. In the 1-shot setting, it obtains 77.7 on HMDB51 and 91.3 on Kinetics-100, exceeding CLIP-FSAR by 1.9 and 1.6 points, respectively. In the 5-shot setting, MuST reaches 89.0 and 97.5, corresponding to improvements of 1.3 and 2.5 points over

Table 1. Comparison under the 5-way protocol. Results for competing methods are taken from their original papers using the same dataset splits.

Method	Backbone	HMDB51		Kinetics-100	
		1-shot	5-shot	1-shot	5-shot
HyRSM [24]	ResNet-50	60.3	76.0	73.7	86.1
MoLo [23]	ResNet-50	60.8	77.4	74.0	85.6
TEAM [6]	ViT-B	70.9	85.5	83.3	92.9
CLIP-FSAR [22]	CLIP ViT-B/16	75.8	87.7	89.7	95.0
MuST (ours)	CLIP ViT-B/16	77.7	89.0	91.3	97.5

Table 2. Ablation of the sub-text pyramid and span-aligned temporal prior under the 5-way 1-shot setting. The random control retains the same span widths and prior strength as box, but randomly permutes the span centres within each level.

Granularity	M	Span prior	HMDB51	Kinetics-100
atomic	4	off	73.9 ± 0.36	89.4 ± 0.19
atomic	4	box	74.4 ± 0.41	89.6 ± 0.21
pyramid	10	off	75.8 ± 0.43	90.4 ± 0.26
pyramid	10	box	77.7 ± 0.34	91.3 ± 0.16
pyramid	10	random	76.1 ± 0.47	90.5 ± 0.27

CLIP-FSAR. MuST also outperforms TEAM by 6.8/3.5 points on HMDB51 and 8.0/4.6 points on Kinetics-100 under the 1-shot/5-shot settings. Among the methods in the table, CLIP-FSAR provides the closest comparison because it uses the same CLIP ViT-B/16 backbone.

4.3 Experimental Results

We further examine MuST through four research questions concerning component contributions, temporal-context modelling, temporal granularity, and sub-text catalogue construction.

RQ1: How do the multi-granularity sub-text pyramid and span-aligned temporal prior contribute to performance?

Table 2 evaluates the two proposed components. Without the temporal prior, replacing the atomic representation with the pyramid improves accuracy by 1.9 points on HMDB51 and 1.0 point on Kinetics-100. The underlying four atomic phases remain unchanged; the pyramid adds their contiguous combinations, increasing M from 4 to 10. Applying the aligned prior to the pyramid produces further gains of 1.9 and 0.9 points, respectively. Together, the two components improve upon the atomic+off configuration by 3.8 points on HMDB51 and 1.9 points on Kinetics-100. Randomly permuting the prior centres reduces performance by 1.6 and 0.8 points relative to the aligned prior. Its differences from pyramid+off are small relative to the reported confidence intervals, providing no clear evidence that an arbitrary temporal constraint is beneficial. The aligned prior gives gains of only 0.5/0.2 points with atomic sub-texts, compared with 1.9/0.9 points with the pyramid. The prior is therefore more effective when used with multi-granularity spans.

RQ2: How does the temporal-context module affect performance?

Table 3 examines the contribution of the temporal-context module. Removing the motion Transformer from the complete configuration reduces accuracy by 1.3 points on HMDB51 and 1.1 points on Kinetics-100. A three-frame convolution recovers 0.6 points on both datasets but remains below the Transformer. On the atomic baseline, the Transformer provides smaller gains of 0.7 and 0.5 points. With temporal context

Table 3. Effect of the temporal-context module under the 5-way 1-shot setting. Rows one and four reproduce the corresponding configurations from Table 2.

Configuration	Temporal context	HMDB51	Kinetics-100
pyramid+box	transformer	77.7 ± 0.34	91.3 ± 0.16
pyramid+box	conv	77.0 ± 0.37	90.8 ± 0.21
pyramid+box	off	76.4 ± 0.41	90.2 ± 0.28
atomic+off	transformer	73.9 ± 0.45	89.4 ± 0.17
atomic+off	off	73.2 ± 0.36	88.9 ± 0.25

Table 4. Granularity sweep on HMDB51 under the 5-way 1-shot setting, with the temporal prior disabled. global uses the class prompt, whereas full concatenates all N phases into one sentence.

Granularity	M	Span lengths retained	Accuracy
global	1	—	72.0 ± 0.40
full	1	$ \omega = 4$	73.1 ± 0.37
atomic	4	$ \omega = 1$	73.9 ± 0.50
atomic_full	5	$ \omega \in \{1, 4\}$	74.7 ± 0.46
pairs	7	$ \omega \in \{1, 2\}$	75.3 ± 0.55
pyramid	10	all	75.8 ± 0.43

disabled, the pyramid+box configuration still exceeds atomic+off by 3.2 points on HMDB51 and 1.3 points on Kinetics-100. The gains from the pyramid and aligned prior therefore remain when the motion Transformer is removed.

RQ3: How does temporal granularity affect performance?

Table 4 compares different span lengths with the temporal prior disabled. Concatenating all N phases into one sentence (full) improves upon the global class prompt by 1.1 points, but remains 2.7 points below the complete pyramid. The full configuration produces one representation for the entire action, whereas the pyramid represents and matches each span separately. Starting from the atomic representation, adding the full-action span improves accuracy by 0.8 points, while adding pairwise spans improves it by 1.4 points. Retaining all span lengths gives the highest accuracy of 75.8 ± 0.43 .

RQ4: How sensitive is MuST to the sub-text catalogue?

Figure 3 examines three aspects of the sub-text catalogue. At $N = 4$, the results obtained with the three tested generators differ by at most 0.7 points, while the aligned prior provides gains between 1.7 and 1.9 points. These results indicate limited sensitivity to the generator within the tested set. Removing the temporal-order requirement changes the result from 75.8 to 75.4 when the prior is disabled. When the prior is enabled, however, its gain decreases from 1.9 to 0.3 points. Temporal ordering therefore has a larger effect on the span-aligned prior than on the pyramid alone. Results remain similar for $N = 3$ and $N = 4$, whereas the gain from the prior decreases to 1.0 point at $N = 5$. Since only $T = 8$ frames are sampled, increasing N beyond four leaves fewer than two

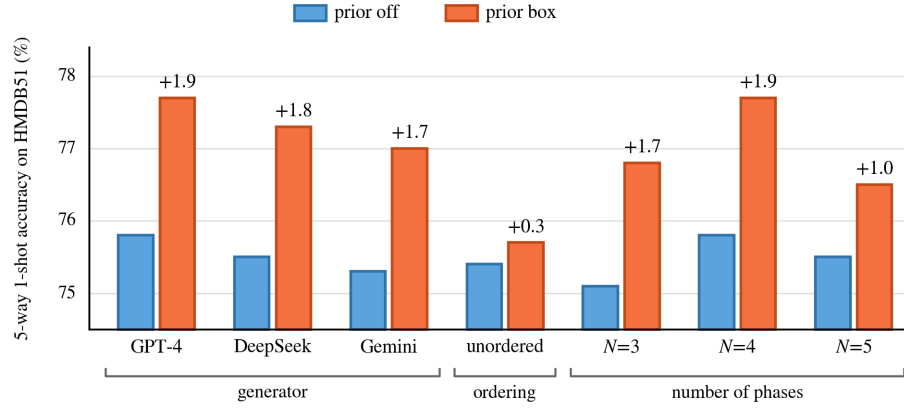


Fig. 3. Sensitivity to the sub-text catalogue on HMDB51 under the 5-way 1-shot setting with pyramid granularity. Each pair of bars reports accuracy with the prior disabled and enabled; Δ denotes the resulting gain. The three blocks compare text generators, temporal ordering, and the number of atomic phases N . The axis begins at 74.5%.

frames per atomic phase on average and may reduce the precision of the temporal prior. We therefore use $N = 4$ in the main experiments.

5 Conclusion

We introduced MuST, a few-shot action recognition framework that represents each action class using ordered sub-texts at multiple temporal granularities. Given N LLM-generated phases, MuST constructs all contiguous spans to form $M = N(N + 1)/2$ sub-text queries. Each query is associated with an expected temporal region, which provides a span-aligned prior for cross-attention between the textual query and video frames. The sub-text queries are encoded once using a frozen text encoder, without updating its pretrained parameters. This design combines fine-grained action phases, longer temporal transitions, and the full action sequence within a unified class representation.

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